ANSWER KEY — Integration Lesson 22 Test: Sigma Notation & Riemann Sums

For the grader · full working shown · method credit for correct setup with small arithmetic slips · justification questions require the reasoning, not just "over" or "under"

1. Evaluate: $\Sigma_{k=1}^{6}$ k.

> **Work:** 1 + 2 + 3 + 4 + 5 + 6 = 21 (or Gauss: 6·7/2 = 21). **Answer: 21.**

2. Evaluate: $\Sigma_{k=1}^{3}$ $(k^2 + 1)$.

> **Work:** k = 1 → 2; k = 2 → 5; k = 3 → 10. Sum: 2 + 5 + 10 = 17. **Answer: 17.** (Classic error: adding the "+1" only once, getting 14 + 1 = 15 — the +1 belongs to EVERY term.)

3. Write in sigma notation: 3 + 6 + 9 + 12.

> **Work:** terms are 3·1, 3·2, 3·3, 3·4 → expression 3k, k from 1 to 4. **Answer: $\Sigma_{k=1}^{4}$ 3k.** (Also accept equivalent forms, e.g. $\Sigma_{k=3}^{6}$… only if it truly generates 3,6,9,12 — $\Sigma_{k=1}^{4}$ 3k is the expected answer; any index letter is fine.)

4. $f(x) = x^2$ on [0, 3], n = 3: LEFT and RIGHT sums.

> **Work:** Δx = (3−0)/3 = 1; edges 0, 1, 2, 3; heights f = 0, 1, 4, 9. Left = 1·(0 + 1 + 4) = 5. Right = 1·(1 + 4 + 9) = 14. **Answer: Left = 5, Right = 14.** (True value 9, for reference. Classic error: using all four heights in one sum — each sum uses exactly n = 3 heights.)

5. $f(x) = x^2$ on [0, 2], n = 2: MIDPOINT sum.

> **Work:** Δx = (2−0)/2 = 1; slices [0,1] and [1,2]; midpoints 0.5 and 1.5. Sum = $1·(0.5^2 + 1.5^2)$ = 0.25 + 2.25 = 2.5. **Answer: 2.5.** (True value 8/3 ≈ 2.67 — midpoint gets close with only two rectangles. Classic error: sampling at edges 0 and 1 or 1 and 2 instead of midpoints.)

6. t (min): 0, 4, 8, 12 · R(t) (L/min): 6, 9, 7, 5 — LEFT sum for total liters.

> **Work:** equal widths of 4 min; left heights 6, 9, 7 (last reading unused). Left = 4·(6 + 9 + 7) = 4·22 = 88. Units: (L/min)·min = L. **Answer: 88 liters.** (Classic error: including the final reading 5 → 4·27 = 108; a left sum never uses the last table entry.)

7. t (s): 0, 1, 4, 6, 10 · v(t) (m/s): 8, 6, 5, 7, 4 — (a) left sum, (b) right sum on [0, 10].

> **Work:** widths: 1−0 = 1, 4−1 = 3, 6−4 = 2, 10−6 = 4. (a) Left heights 8, 6, 5, 7: 8·1 + 6·3 + 5·2 + 7·4 = 8 + 18 + 10 + 28 = 64. (b) Right heights 6, 5, 7, 4: 6·1 + 5·3 + 7·2 + 4·4 = 6 + 15 + 14 + 16 = 51. **Answer: (a) 64 meters (b) 51 meters.** (Classic error: assuming a common Δx like 2.5 — the widths are unequal and must be computed from the table.)

8. $f(x) = \sqrt{x}$ increasing on [1, 4]: does a RIGHT sum over- or underestimate $\int_1^4 \sqrt{x}\,dx$? Justify.

> **Work:** full-credit sentence: "Because f is increasing on [1, 4], each right-endpoint height is the largest value of f on its subinterval, so every rectangle contains area above the curve; therefore the right Riemann sum overestimates the integral." **Answer: overestimates (with the increasing-function justification).** (Grader note: the bare word "overestimate" with no reasoning earns half credit at most; the justification must connect increasing → right edge is the max height.)

9. R(t) in liters per minute, t in minutes: what does $\int_0^{20} R(t)\,dt$ represent?

> **Work:** integrating a rate gives a total; units: (liters/minute) × minutes = liters. **Answer: the total number of LITERS of juice poured into the vat during the first 20 minutes.** (Full credit needs all three: the quantity (liters), the interval (from t = 0 to t = 20), and the idea of total/accumulated amount. "The area under the curve" alone is not an interpretation.)

10. t (hr): 0, 3, 5, 9 · r(t) (gal/hr): 12, 10, 8, 6 — (a) left sum, (b) right sum, (c) which overestimates? Justify.

> **Work:** widths: 3−0 = 3, 5−3 = 2, 9−5 = 4. (a) Left heights 12, 10, 8: 12·3 + 10·2 + 8·4 = 36 + 20 + 32 = 88. (b) Right heights 10, 8, 6: 10·3 + 8·2 + 6·4 = 30 + 16 + 24 = 70. (c) "Since r is decreasing, each left-endpoint height is the maximum of r on its subinterval, so the left rectangles lie above the curve — the left sum, 88 gallons, is the overestimate." **Answer: (a) 88 gallons (b) 70 gallons (c) the left sum, because r is decreasing.** (Score each part separately; (c) needs the decreasing→left-max reasoning, not just "left".)